

ICHEON, Korea, Republic of

WMO: 470970

Lat: 37.2639N

Lon: 127.4842E

Elev: 260

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	10.3	-10.5	3.2	16.9	-6.9	3.8	17.5	11.1	28.8	9.9	28.2	2.4	270	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.7	91.3	75.7	88.8	74.4	86.3	72.5	78.4	85.9	77.3	84.9	76.1	83.4	4.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	140.3	81.9	75.2	133.6	81.7	73.7	126.8	80.5	42.3	85.8	41.0	84.8	39.8	83.9	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.1	9.5	8.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			0.8	94.8	5.9	2.5	-3.5	96.6	-6.9	98.0	-10.3	99.4	-14.6	101.2
			-0.1	80.2	5.6	2.0	-4.1	81.6	-7.3	82.7	-10.4	83.8	-14.5	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	25.2	31.6	42.0	53.0	64.4	72.6	77.6	78.3	69.0	56.8	43.2	29.0	(6)
(7)		DBStd	19.36	7.74	6.97	6.59	6.19	4.61	3.41	3.69	4.66	4.67	6.19	7.97	7.62	(7)
(8)		HDD50	2487	768	515	263	41	0	0	0	0	16	233	651		(8)
(9)		HDD65	5360	1233	935	712	362	67	1	0	16	263	655	1116		(9)
(10)		CDD50	3829	0	0	16	130	447	679	856	878	569	225	29	0	(10)
(11)		CDD65	1228	0	0	0	1	50	230	391	413	135	8	0	0	(11)
(12)		CDH74	10040	0	0	0	45	585	1703	3174	3725	768	40	0	0	(12)
(13)		CDH80	3096	0	0	0	3	105	443	1017	1407	120	1	0	0	(13)
(14)	Wind	WSAvg	3.0	2.8	3.2	4.0	4.1	3.4	3.0	2.9	2.6	2.2	2.2	2.7	2.7	(14)
(15)	Precipitation	PrecAvg	51.6	1.0	1.1	2.0	3.8	3.9	5.6	13.3	9.9	6.3	1.7	1.9	1.1	(15)
(16)		PrecMax	91.1	2.9	5.0	4.8	10.2	10.5	18.0	27.4	23.4	28.0	8.3	5.7	2.2	(16)
(17)		PrecMin	37.4	0.0	0.0	0.4	0.3	0.9	0.5	3.9	2.8	0.2	0.0	0.2	0.1	(17)
(18)		PrecStd	12.2	0.9	1.2	1.1	2.7	2.4	4.4	6.4	5.5	5.9	1.7	1.4	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	55.2	69.1	79.4	86.1	89.0	93.4	95.2	86.1	78.2	69.0	50.7	(19)
(20)		0.4%	MCWB	39.4	45.7	53.2	56.8	61.6	66.5	76.2	77.8	72.7	65.4	58.6	44.8	(20)
(21)		2%	DB	42.9	51.2	64.2	74.2	82.8	86.4	90.3	92.6	83.1	74.1	62.6	46.6	(21)
(22)		2%	MCWB	35.7	41.7	49.5	54.3	60.6	66.7	75.2	77.1	68.8	61.4	53.5	40.7	(22)
(23)		5%	DB	39.5	47.1	59.4	70.6	80.1	84.7	87.9	90.0	81.0	70.9	59.0	43.7	(23)
(24)		5%	MCWB	33.5	38.4	46.6	52.5	60.1	66.5	74.7	76.3	67.3	59.2	51.4	38.6	(24)
(25)		10%	DB	36.5	43.0	55.3	66.7	77.2	82.7	85.6	87.5	78.7	68.1	55.8	40.4	(25)
(26)		10%	MCWB	31.1	35.6	44.3	50.9	58.8	66.0	73.8	75.1	66.4	58.2	49.7	35.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	50.3	55.3	59.4	68.5	72.9	79.5	80.7	76.6	69.0	59.9	47.3	(27)
(28)		0.4%	MCDWB	46.3	53.5	63.0	73.1	71.5	79.6	85.8	89.3	82.6	73.5	65.8	49.1	(28)
(29)		2%	WB	37.0	42.6	51.9	57.3	64.6	71.1	77.9	79.1	73.4	64.9	55.7	42.0	(29)
(30)		2%	MCDWB	41.5	49.6	60.5	68.9	75.8	79.1	84.3	87.3	77.5	71.7	61.3	45.1	(30)
(31)		5%	WB	34.2	39.1	48.8	55.0	62.8	69.9	76.9	78.0	71.1	61.6	52.9	39.0	(31)
(32)		5%	MCDWB	38.2	45.9	57.3	65.7	74.1	78.4	84.3	86.0	77.1	67.5	57.7	42.9	(32)
(33)		10%	WB	31.8	36.6	45.3	52.9	61.3	68.6	75.7	76.9	69.1	59.1	50.2	35.7	(33)
(34)		10%	MCDWB	36.1	42.4	53.8	63.0	72.3	77.7	82.7	84.6	75.3	65.9	54.7	39.8	(34)
(35)	Mean Daily Temperature Range		MDBR	17.7	18.2	20.9	22.7	22.5	18.6	13.2	14.7	18.2	21.8	17.8	15.9	(35)
(36)		5% DB	MCDBR	18.3	23.6	27.6	30.3	28.3	22.5	17.4	18.4	21.3	24.0	21.1	17.7	(36)
(37)			MCWBR	12.6	15.0	14.8	13.2	9.8	7.0	5.6	6.2	8.6	11.6	12.6	12.9	(37)
(38)			MCDBR	16.5	21.0	23.6	23.7	20.8	15.7	13.0	15.6	14.7	17.9	18.0	16.0	(38)
(39)		5% WB	MCWBR	12.0	14.5	14.0	12.2	9.5	6.3	5.1	6.2	6.8	10.2	11.8	13.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.425	0.476	0.561	0.599	0.610	0.677	0.643	0.568	0.492	0.480	0.462	0.420	(40)	
(41)		taud	1.971	1.881	1.731	1.693	1.714	1.638	1.764	1.948	2.088	1.996	1.982	2.005	(41)	
(42)		Ebn at Noon	226	230	223	224	225	210	216	230	241	227	212	217	(42)	
(43)		Edh at Noon	43	54	68	75	75	80	70	57	47	47	43	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	785	1027	1328	1547	1746	1621	1266	1335	1271	1121	778	700	(44)	
(45)		RadStd	47	82	138	120	153	144	184	192	146	83	81	51	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+88	(47)

Nomenclature: See separate page